8 EXAM PRACTICE

1 The equation for a reaction is:

$Ca(OH)_2 + 2HNO_3 \rightarrow Ca(NO_3)_2 + 2H_2O$

Which description is correct for this reaction?

A oxidation only

B reduction only

C oxidation and reduction

D neither oxidation nor reduction [1]

(Total for Question 1 = 1 mark)

2 What is the oxidation number of chlorine in the compound $KClO_4$?

A −1

B +1

C +4

D +7 [1]

(Total for Question 2 = 1 mark)

3 Which equation represents a disproportionation reaction?

A $2LiOH + H_2SO_4 \rightarrow Li_2SO_4 + 2H_2O$

B $2H_2S + SO_2 \rightarrow 3S + 2H_2O$

C $2NO_2 + H_2O \rightarrow HNO_2 + HNO_3$

D $KF + H_2SO_4 \rightarrow KHSO_4 + HF^-$ [1]

(Total for Question 3 = 1 mark)

4 Which equation represents the process that occurs when the second ionisation energy of strontium is measured?

A $Sr(s) \rightarrow Sr^+(g) + e^-$

B $Sr^+(s) \rightarrow Sr^{2+}(g) + e^-$

C $Sr(g) \rightarrow Sr^+(g) + e^-$

D $Sr^+(g) \rightarrow Sr^{2+}(g) + e^-$ [1]

(Total for Question 3 = 1 mark)

5 A combination of reagents that can be used to test for sulfate ions in solution is $Ba(NO_3)_2$ and HNO_3.

Which other combination of reagents can also be used in this test?

A BaO + NaOH

B $BaCO_3$ + NaOH

C $BaSO_4$ + HCl

D $BaCl_2$ + HCl [1]

(Total for Question 5 = 1 mark)

6 A mass of 830 mg of KOH is dissolved in water to make 150 cm^3 of solution.

What is the concentration of hydroxide ions in this solution?

A 0.0148 g dm^{-3}

B 0.0986 mol dm^{-3}

C 0.325 mol dm^{-3}

D 5.53 mg dm^{-3} [1]

(Total for Question 6 = 1 mark)

7 Most chlorine-based bleaches contain the ClO^- ion.

(a) (i) State the oxidation number of the chlorine in the ClO^- ion. [1]

(ii) Give the systematic name of the ClO^- ion. [1]

(b) The ClO^- ion is produced when chlorine gas is reacted with water.

Write an ionic equation for this reaction and explain why chlorine is said to undergo 'disproportionation'. [3]

(c) Chlorine forms another ion with the formula ClO_3^-. This ion exists in the compound $KClO_3$.

$KClO_3$ decomposes on heating according to the equation:

$4KClO_3(s) \rightarrow 3KClO_4(s) + KCl(s)$

Comment on the change in oxidation number of the chlorine in this reaction. [3]

(Total for Question 7 = 8 marks)

8 Bromine is present in sea water as aqueous bromide ions, $Br^-(aq)$. Bromine can be extracted from sea water by adding chlorine. The ionic equation for the reaction is:

$Cl_2(aq) + 2Br^-(aq) \rightarrow 2Cl^-(aq) + Br_2(aq)$

(a) (i) State what is meant by oxidation and reduction in terms of electron transfer. [1]

(ii) State which species is acting as an oxidising agent. Justify your answer. [2]

(b) The chlorine required for extraction of bromine is manufactured from a concentrated solution of sodium chloride using electrolysis.

The ionic half-equations for the reactions at the electrodes are:

At the anode: $2Cl^-(aq) \rightarrow Cl_2(aq) + 2e^-$

At the cathode: $2H_2O(l) + 2e^- \rightarrow 2OH^-(aq) + H_2(g)$

Use these two half-equations to produce an overall ionic equation for the reaction taking place during the electrolysis. [2]

(Total for Question 8 = 5 marks)

9 This question is about the chemistry of the elements in Groups 1 and 2 of the Periodic Table.

(a) When magnesium burns in air it forms a mixture of magnesium oxide and magnesium nitride.

Give the formula of both magnesium oxide and magnesium nitride. [2]

(b) (i) Write an equation for the reaction between calcium and water. Include state symbols. [2]

(ii) Explain how the reactivity of the Group 2 metals with water changes down the group. [3]

(iii) State how the solubility of the hydroxides of the Group 2 metals changes down the group. [1]

(c) Explain the trend in thermal stability of the carbonates of the Group 2 metals. [3]

(d) The nitrates of most of the Group 1 metals decompose on heating as shown in the equation:

$$MNO_3 \rightarrow MNO_2 + \tfrac{1}{2}O_2$$

where M represents a Group 1 metal.

However, lithium nitrate decomposes further.

Write an equation for the thermal decomposition of lithium nitrate. State symbols are not required. [2]

(Total for Question 9 = 13 marks)

10 This question is about halides.

(a) Group 1 halides react with concentrated sulfuric acid in different ways.

Sodium chloride reacts to give hydrogen chloride as the only gaseous product.

Sodium bromide reacts to give both hydrogen bromide and bromine as gaseous products.

(i) Write an equation for the reaction of both sodium chloride and sodium bromide with concentrated sulfuric acid. [2]

(ii) Explain why the two reactions produce different gaseous products. [3]

(iii) Describe how ammonia gas can be used to show that hydrogen chloride has been given off in the reaction between sodium chloride and concentrated sulfuric acid. [2]

(b) Hydrogen halides react with water.

(i) Write an equation for the reaction between hydrogen bromide and water. [1]

(ii) Explain the effect that the solution formed has on methyl orange indicator. [3]

(c) Describe how you would show that an aqueous solution contains both chloride and iodide ions. [4]

(Total for Question 10 = 15 marks)

11 Radium (Ra) and astatine (At) are radioactive elements in Groups 2 and 7 respectively in the Periodic Table.

(a) Predict the appearance at room temperature of

(i) radium [1]

(ii) astatine. [2]

(b) Predict the reactivity of astatine compared to that of iodine. Justify your answer. [3]

(c) Write an equation for the reaction of astatine with hydrogen. State symbols are not required. [1]

(d) Write an equation for the reaction between radium and chlorine. State symbols are not required. [1]

(e) The product of the reaction between radium and water, and the pH of the resulting solution, are most likely to be:

A radium oxide and 1

B radium oxide and 14

C radium hydroxide and 1

D radium hydroxide and 12 [1]

(Total for Question 11 = 9 marks)

12 This question is about some compounds of the elements in Group 2 of the Periodic Table.

(a) Water was slowly added to a solid lump of calcium oxide. The lump got very hot and broke apart to form another white solid.

Excess water was added to this white solid and the mixture was filtered to produce a colourless solution. The solution turned milky when carbon dioxide was bubbled through it.

(i) State why the lump of calcium oxide got very hot when water was added. [1]

(ii) Name the colourless solution formed when an excess of water was added to the white solid, and predict its pH. [2]

(iii) Write an equation for the reaction between this colourless solution and carbon dioxide. Include state symbols. [2]

(b) Calcium oxide can be formed in several ways.

One way is to heat calcium in oxygen. Another way is to heat calcium nitrate.

(i) Write an equation for each method of forming calcium oxide. State symbols are not required. [2]

(ii) Explain the trend in decomposition temperatures of the Group 2 nitrates. [4]

(Total for Question 11 = 11 marks)